Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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[single]acro sdsshort=SDS, long=spoken dialogue system, long-plural=s, short-plural=s yrrsds-short=YRRSDS, long=Young Researchers' Roundtable on Spoken Dialog Systems, long-plural=, short-plural=

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1 Research interests

The following instructions are directed to authors of position papers submitted for participation in the *yrrsds*. All authors are required to adhere to these specifications. Since the submissions will be published in an electronic format, the authors are required to provide only a Portable Document Format (PDF) version of their papers. *Before sending in your submission, please test your PDF by printing it from a computer different from the one where it was created.* Templates for both LaTeX and Word documents are available at `http://yrrsds.org`.

Submitted position papers should not exceed two pages. The papers will be composed of three main sections, followed by a section for references, and a biographical sketch. The titles of the 3 main sections **must remain**: Research interests, sds research, and Suggested topics for discussion. **These section titles should NOT be changed, but subsection titles should**.

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1.1 Name of first topic of interest

Authors will provide an overview of their work in subsections of sec:research$_i$*nterests−−onepertopic.Theycanbearrangedl*
leveldescriptionofspecificquestionsbeingaddressed, *ofthemethodsused*, *andoftheresultsobtainedsofar*(*ifapplicable*).*1*

1.2 Name of second topic of interest

The research interests in sec:research$_i$*nterestscanbefurtherdividedintosubsections*, *asfoundappropriatebytheauthor*.

2 sds research

In this section, the authors will briefly present their opinions and point of views of current and future sds research in general. These could include answers to questions like:

- Where do you think the field of dialogue research will be in 5 to 10 years?
- What do you think this generation of young researchers could accomplish in that time?
- What kind of questions need to be investigated to get the field to that point?

- What are the most important things for users of sds
- Is there a difference between sds research in academia and industry?
- Will sds be more widely used in the future? How? In what scenarios?
- ...

3 Suggested topics for discussion

Here, authors will suggest three topics for discussion in the discussion panels during the event. As an example, here are some of the discussion topics discussed in previous workshops:

- Statistical methods and user simulations: Best practices for building and evaluation user simulations, challenges in using reinforcement learning for building systems.
- Evaluation: Global competition, universal metrics for comparing disparate systems.
- Next killer application: Speech for web browsing, in-car systems, mobile devices, health-care, interface to robots.
- Dialogue and Question-Answering: Mutual benefits, user interaction.
- Affect and other paralinguistic phenomena: Emotion detection and handling, turn taking, response timing.
- Rapid development and standardization.
- ...

It is recommended to suggest topics, on which the author has knowledge, but also topics that they find interesting and relevant to the young community.

Biographical sketch

myphoto.png

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Finally, authors will provide a brief biographical sketch (1 or 2 paragraphs). The bio will be written in third person, and can include a brief description of the author's academic and professional history, career plans, as well as extra-curricular interests. Authors are encouraged to include a photo (width = 1.2", height = 1.25"), to make it easy to recognize participants at the event This however is not a requirement (if the author does not wish to include a photo, he or she should simply delete the image at the beginning of this paragraph).

***** End of position paper template. Delete the instructions below after reading them *****

4 Credits

This document has been adapted from the instructions for the yrrsds 2005–2020. In turn, those documents have been adapted from the instructions for the NAACL-2004 proceedings, which again was adapted from the instructions for the ACL-2002 proceedings by Dekang Lin and Eugene Charniak. It was adapted in turn from the ACL-02 proceedings by Norbert Reithinger, Giorgio Satta, and Roberto Zamparelli. The ACL-01 instructions were elaborated from similar documents used for previous editions of the ACL and EACL annual meetings. Those versions were written by several people, including John Chen, Henry S. Thompson and Donald Walker. Additional elements were taken from the formatting instructions of the International Joint Conference on Artificial Intelligence.

5 General Instructions

Manuscripts must be in single-spaced, two-column, single-page format. Start all pages directly under the top margin. If the paper is produced by a printer, make sure that the quality of the output is dark enough to photocopy well. It may be necessary to have your laser printer adjusted for this purpose. Papers that are too faint to reproduce well may not be included.
Do not print page numbers on the manuscript.
The maximum length of a manuscript is two 2 pages.

6 Format of electronic manuscript

Only manuscripts in Portable Document Format (PDF) will be accepted. It is **strongly** recommended to only use LaTeXfor producing the manuscript, since the output of other programs (like Microsoft Word) might not be compatible. Moreover, some word processor may generate very large postscript/PDF files, where each page is rendered as an image. Such images may shown poorly on a screen. Please make sure that the PDF is shown correctly in a PDF standard viewer. It is also recommended to test the file on a computer different from the one on which it was produced.

For uniformity reasons, **Times Roman** font should be used. In LaTeX2e this is accomplished by putting by loading the *times* and *latexsym* packages in the preamble.

6.1 Layout

Manuscripts should have two columns to a page, in the manner this template is printed. The exact dimensions for a page on A4 paper are:

Text and subsection font size should be set to 10 points; Sections font size to 12 points; And the title font size to 15 points.

6.2 Headings

Section and subsection headings should be numbered using Arabic numerals to facilitate cross references. Subsection number should be separated with a dot from the section number. Do not

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Left and right margins	2.4
Top margin	2.4
Bottom margin	5.0
Columns width	7.9
Gap between columns	0.4

number `subsubsections`.

6.3 Citations

Citations should follow the Guidelines for Formatting Submissions to *Computational Linguistics*. That is, citations within the text appear in parentheses gusfield1997algorithms. Or only the year in brackets if the author's name appears in the text itself, as in the case of gusfield1997algorithms. Append lowercase letters to the year in cases of ambiguities. Treat double authors as in aho1972theory, but write as in chandra1981embedded when more than two authors are involved. Collapse multiple citations wagner1974string, chandra1981embedded.

6.4 References

Gather the full set of references together under the heading **References**. Place this section just before the biographical sketch. Arrange the references alphabetically by first author. Provide as complete a citation as possible, using a consistent format, such as the one for *Computational Linguistics* or the one in the *Publication Manual of the American Psychological Association* american1994publication. Use of full names for authors rather than initials is preferred. A list of abbreviations for common computer science journals can be found in the ACM *Computing Reviews* american1994publication. It is recommended to use an external *.bib* file for managing the references, but putting the entries directly in the document are acceptable as well.

6.5 Acknowledgments

Acknowledgments section, if needed, should go as a last section right before the references. Do not number the Acknowledgments section.

6.6 Footnotes

Put footnotes at the bottom of the page. They should be numbered or referred. [1] Footnotes should be separated from the text by a line. [2]

[1] This is how a footnote should appear.

[2] Note the line separating the footnotes from the text.

6.7 Graphics

Place figures, tables, and photographs in the paper near where they are first discussed, rather than at the end. Wide illustrations may run across both columns, using an `includefigure*` block.

6.8 Captions

Provide a caption for every illustration; number each one sequentially in the form: Figure 1. Caption of the Figure., Table 1. Caption of the Table.. Caption of figures should come below them, and captions of tables above.